Chemistry · Unit 1A · Level 1

Charges & Static



TODAY'S PLAN

- ⚡ What is lightning?
- ⚛️ Tiny atoms
- ➖ Electrons move
- 🎈 Static

Review · 5 minutes

Quick Review



⚡ Lightning is a **big spark**.

⛈️ It comes from **storm clouds**.

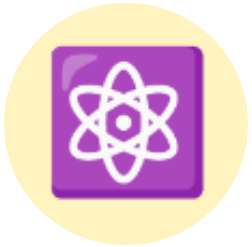
👉 Point: lightning is 🔥 hot or 🧊 cold?

Words to Know

Words to Know



Lightning
a big spark in a storm



Atom
a tiny piece of stuff



Proton
plus (+) part of an atom



Electron
minus (-) part that moves

Learn

Lightning

⚡ **Lightning** is a giant **spark**.

🌩️ It jumps from a **cloud** to the **ground**.

🔥 It is very **hot** and very **fast**.



Check 1 of 4



Where does lightning come from?

Point to the picture!

The sun

Storm cloud

✅ Answer · Card 1 of 4



Storm cloud

Lightning comes from storm clouds.

Learn

Everything Is Made of Atoms



⚛️ Everything is made of tiny **atoms**.

➕ **Protons** are plus.

➖ **Electrons** are minus.

⚪ **Neutrons** have no charge.

Check 2 of 4



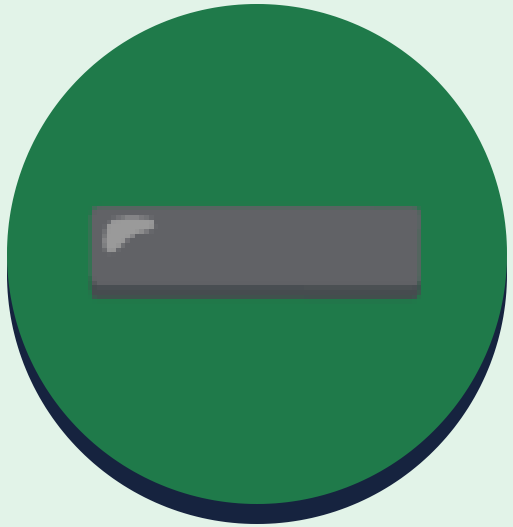
Which part of the atom is minus (–)?

Point to the picture!

➖ Electron

➕ Proton

✅ Answer · Card 2 of 4



Electron

Electrons are negative (−).

Learn

Rubbing Moves Electrons

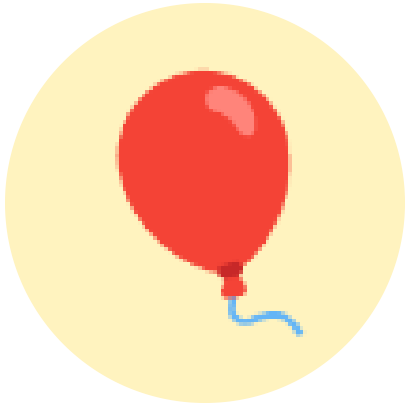
🎈 Rub a **balloon** on **hair**.

➖ **Electrons** move to the balloon.

⚡ Now the balloon has a **static charge**.



🃏 Check 3 of 4



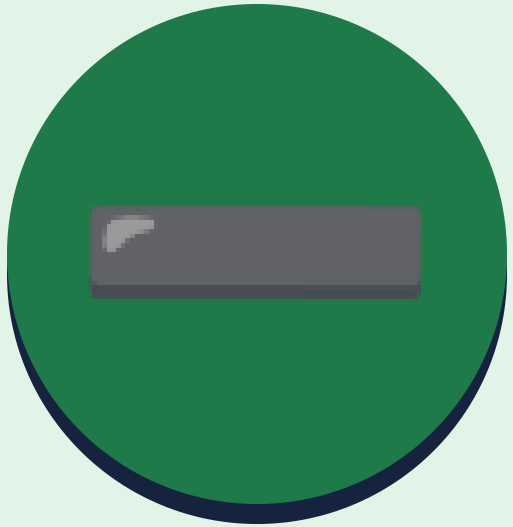
What moves when you rub a balloon on hair?

Point to the picture!

➖ Electrons

💧 Water

✅ Answer · Card 3 of 4



Electrons

Only electrons move.

Steps

Sparks



1. 🎈 Charge builds up.
2. ⚡ The charge jumps.
3. ✨ We see a spark!
4. 📣 Sometimes we hear a snap.

🃏 Check 4 of 4



A spark is...

Point to the picture!

💨 Wind

⚡ Moving charge

✅ Answer · Card 4 of 4



Moving charge

A spark is charge jumping across air.

Learn

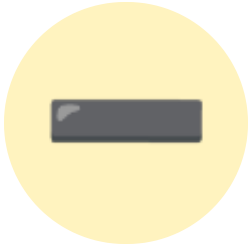
Lightning Is a Big Spark



Small spark
balloon or doorknob



Big spark
lightning



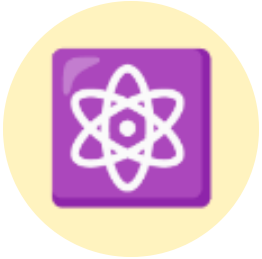
Same cause
electrons move



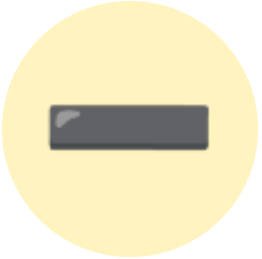
Light + sound
flash and snap or thunder

Wrap-Up

Recap



Atoms
tiny pieces



Electrons
move



Static
charge builds



Lightning
big spark

Wrap-Up

Great Job!



⚛️ Everything is made of atoms.

➖ Electrons move when we rub.

⚡ Lightning is a big spark.